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Pd-catalyzed Fukuyama cross-coupling of secondary organozinc reagents for the direct synthesis of unsymmetrical ketones

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ABSTRACT

The coupling of acyl electrophiles with organometallic reagents represents a convergent route toward complex and versatile ketone products. Despite the mild conditions and high functional group tolerance, the cross-coupling of carboxylic acid derivatives, such as thioesters, and secondary organometallic reagents is an underdeveloped transformation. Herein, we disclose a convenient and efficient protocol for the Pd-catalyzed Fukuyama cross-coupling of secondary organozinc reagents with thioester electrophiles. Under these mild conditions, a range of thioesters possessing sensitive functional groups can be coupled with either activated or unactivated secondary organozinc halides in good yields. This method was expanded to include an acid chloride substrate, generating an aryl alkyl ketone in high yield. In addition, a modest dynamic kinetic resolution of the organozinc reagent can be achieved using chiral phosphoramidite ligands to furnish enantioenriched ketone products.

1. Introduction

Transition metal-catalyzed cross-coupling reactions have emerged as a transformative methodology in the field of organic synthesis, enabling the rapid generation of complex molecules under mild, selective, and catalytic conditions. While initial reports focused on C(sp^2)–C(sp^2) couplings, the burgeoning field of C(sp^3) bond-forming processes has rapidly expanded in the past decade.[1] In a seminal report, Hayashi and co-workers found that Pd complexes supported by bulky, bidentate phosphines with large bite angles (e.g., 1,1′-bis(diphenylphosphino)ferrocene) (dppf) can catalyze the cross-coupling between aryl or vinyl halides and primary or secondary organomagnesium and organozinc reagents.[2] In 2003, Fu and co-workers demonstrated that similar bulky, electron-rich phosphines enable a general Pd-catalyzed Negishi coupling of sp^3-hybridized electrophiles with alkyl, alkenyl, and aryl nucleophiles.[3] In the decade that has followed, steady advances have been made in Pd- and Ni-catalyzed cross-coupling reactions of both secondary C(sp^3) organometallic reagents and secondary C(sp^3) electrophiles, giving rise to a wide variety of products now accessible via simple alkyl building blocks.[4–6]

One transformation that remains underdeveloped is the transition metal-catalyzed cross-coupling reaction between carboxylic acid derivatives and C(sp^3) organometallic reagents to prepare ketones, despite the fact that the Pd-catalyzed reaction between acid chlorides and alkyl tin reagents was first reported over thirty years ago.[7] In 1998, Fukuyama and co-workers reported the Pd-catalyzed coupling of thioesters and primary alkyl organozinc or hydride reagents, generating ketone or aldehyde products, respectively.[8] A number of functional groups are tolerated due to the relative stability of both coupling components. Seki and co-workers[9] have since developed both heterogeneous and phosphine-free Pd-catalyzed Fukuyama couplings, as well as a Ni-catalyzed Fukuyama coupling. Despite these advances, the use of secondary C(sp^3) organometallic reagents still proves challenging. While α,α-disubstituted ketones can be prepared by direct attack of a variety of strongly nucleophilic organometal species onto acyl electrophiles, such protocols are frequently accompanied by over-addition products due to the electrophilicity of the newly formed ketone.[10] Specialized acyl derivatives, such as Weinreb amides, can minimize over-addition; however, these electrophiles require the use of organolithium or Grignard reagents, which suffer from poor functional group tolerance.[11] Recent strategies to eliminate the need for alkyl organometallic reagents in ketone synthesis include reverse-polarity cross-couplings[12] and reductive cross-couplings.[13]

Although substantial progress has been made on the transition metal-catalyzed cross-coupling between acyl electrophiles and primary alkyl organometallic reagents, there are far fewer reports describing the coupling of secondary alkyl organometallics. In an early study, Harada and Oku disclosed the coupling of secondary organozinc reagents with acid chlorides and obtained moderate to good yields of ketone product, although the functional group tolerance of the reaction was not further explored (Fig. 1).[14] Zhang and

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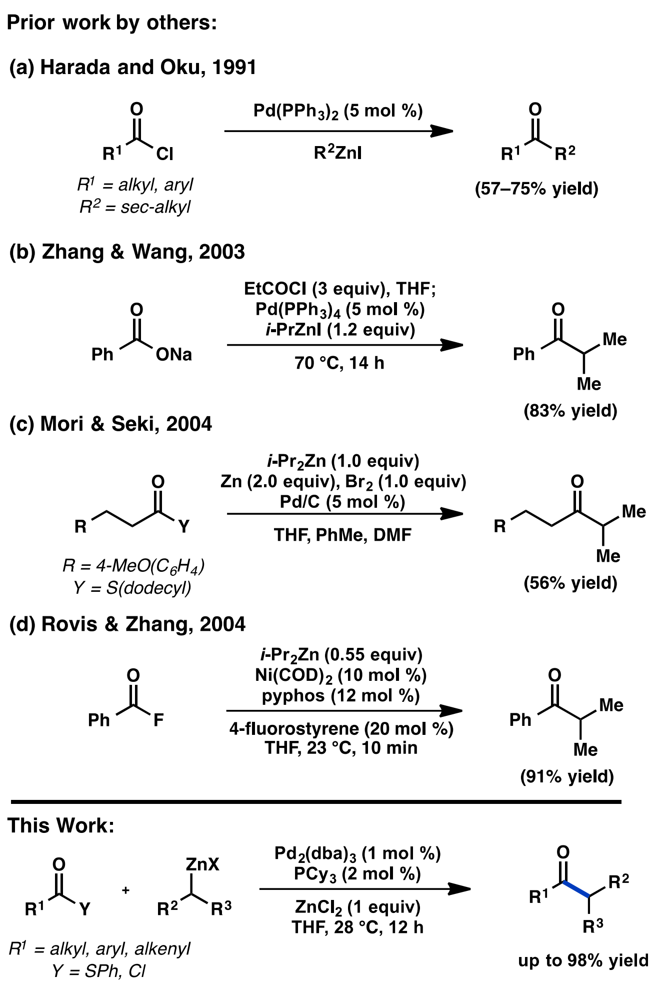


Fig. 1. Acyl cross-coupling reactions of secondary organozinc reagents.

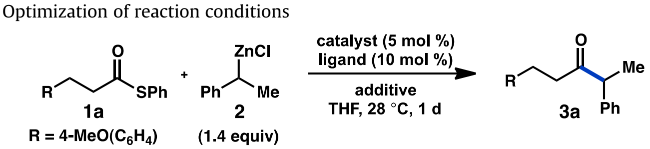
Wang reported a high-yielding reaction between *i*-PrZnI and the mixed anhydride generated from sodium benzoate, however additional substrate scope was not disclosed.[15] Subsequently, Mori and Seki reported a Pd-catalyzed coupling of *i*-Pr_2Zn that proceeds in moderate yield.[16] Rovis and Zhang have also reported a single case of a high-yielding coupling between an acid fluoride and *i*-Pr_2Zn.[17] Finally, the related Liebeskind–Srogl coupling of thioesters and organoboron nucleophiles proceeds only with primary organoboron reagents despite efforts to extend the reaction to secondary reagents.[18] With the objective of expanding the synthetic utility of metal-catalyzed acyl cross-coupling reactions, herein we report a general method for the cross-coupling of secondary organozinc reagents and thioesters.

2. Results and discussion

The scarcity of *sec*-alkylmetal couplings with acyl derivatives is indicative of the inherent difficulties of transition metal-catalyzed reactions involving C(sp^3)-hybridized coupling partners.[1d] In general, alkyl organometallic reagents are prone to β-hydride elimination as well as proto-demetalation under typical cross-coupling reaction conditions. Additionally, many alkyl organometallics suffer from slow transmetalation to the transition metal catalyst. Furthermore, not only is transmetalation troublesome for *sec*-alkylmetals, but reductive elimination from a transition metal is also slow due to σ-donation of the alkyl group to the metal. The sluggishness of reductive elimination enhances side reactions, such as β-hydride elimination and isomerization. Acceleration of the reductive elimination pathway, either through catalyst and ligand tuning or through the use of additives, can prevent this undesired isomerization. Significantly, alkylzinc reagents display relatively broad use in C(sp^3) cross-couplings because of their ability to undergo a facile transmetalation with Pd as compared to other organometallic partners used in cross-coupling chemistry.

We therefore began our studies by evaluating the cross-coupling between thioester **1a** and organozinc **2**. An analysis of metal catalysts revealed that Ni sources performed poorly (Table 1, entries 1 and 2), whereas $PdCl_2$(dppf) was more promising (Table 1, entry 5). A screen of several phosphine ligands revealed that the bulky, monodentate, electron-rich ligand PCy_3 furnished ketone **3a** in 34% yield (entries 6–9). Despite their success in a number of Negishi couplings, the Buchwald ligands SPhos and XPhos showed low reactivity.[19] On the other hand, use of 10 vol % DMF as a polar cosolvent further increased the yield to 52% (entry 10), although additional DMF had no effect on the reaction. Other trialkylphosphines resulted in lower yields of **3a** (entries 11–14). Interestingly, these conditions of $Pd_2(dba)_3$/PCy_3/DMF closely resemble Fu's general conditions ($Pd_2(dba)_3$/$PCyp_3$/NMI) for Negishi cross-couplings of alkyl electrophiles.[3] Notably, no isomerized linear product was observed.

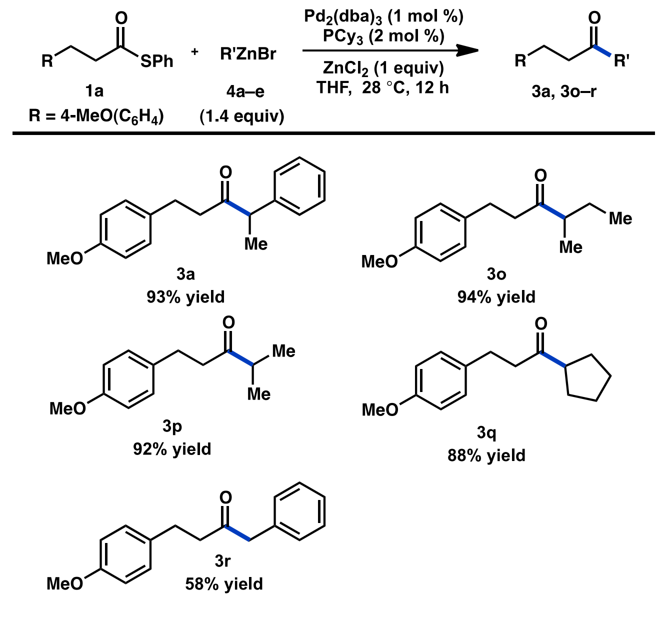
Table 1
Optimization of reaction conditions



Entry	Catalyst	Ligand[a]	Additive	Conversion (%)[b]	Yield[b] (%)
1	$Ni(acac)_2$[c]	—	—	75	10
2	$NiCl_2$(glyme)[c]	—	—	83	14
3	$Pd_2(dba)_3$	—	—	94	13
4	$Pd(OAc)_2$[c]	—	—	75	15
5	$PdCl_2$(dppf)[c]	—	—	88	20
6	$Pd_2(dba)_3$	PPh_3	—	92	16
7	$Pd_2(dba)_3$	PCy_3	—	89	34
8	$Pd_2(dba)_3$	SPhos	—	93	16
9	$Pd_2(dba)_3$	XPhos	—	82	14
10	$Pd_2(dba)_3$	PCy_3	DMF[e]	78	52
11	$Pd_2(dba)_3$	PMe_3	DMF[e]	100	23
12	$Pd_2(dba)_3$	PEt_3	DMF[e]	80	35
13	$Pd_2(dba)_3$	P^nBu_3	DMF[e]	90	32
14	$Pd_2(dba)_3$	P^tBu_3	DMF[e]	88	22
15	$Pd_2(dba)_3$[d]	PCy_3	DMF[e]/$ZnCl_2$[f]	100	83
16	$Pd_2(dba)_3$[d]	PCy_3	$ZnCl_2$[f]	93	89
17	$Pd_2(dba)_3$[d]	dCyb	$ZnCl_2$[f]	49	44

[a] 1:1 Pd:phosphine was used.
[b] Conversion of **1a** and yield of **3a** were calculated by ^1H NMR analysis with internal standard.
[c] 10 mol % was used.
[d] 1 mol % was used.
[e] 10 vol % was added.
[f] 1 equiv was added.

Previous studies of the Fukuyama coupling have noted that addition of inorganic zinc salts may shift the Schlenk equilibrium of the organozinc reagent toward a more reactive organozinc halide and may also activate Pd toward oxidative addition.[9] Gratifyingly, addition of $ZnCl_2$ increased both the conversion and yield of the reaction, while allowing the catalyst loading to be reduced to 1 mol % $Pd_2(dba)_3$ without an appreciable decrease in yield (entry 15). Use of $ZnCl_2$ as the sole additive also obviated the need for DMF, delivering **3a** in 89% yield (entry 16). Replacement of PCy_3 for a bidentate phosphine analog, 1,4-bis(dicyclohexylphosphino)butane (dCyb), resulted in both a lower conversion of **1a** and a lower yield of **3a**.



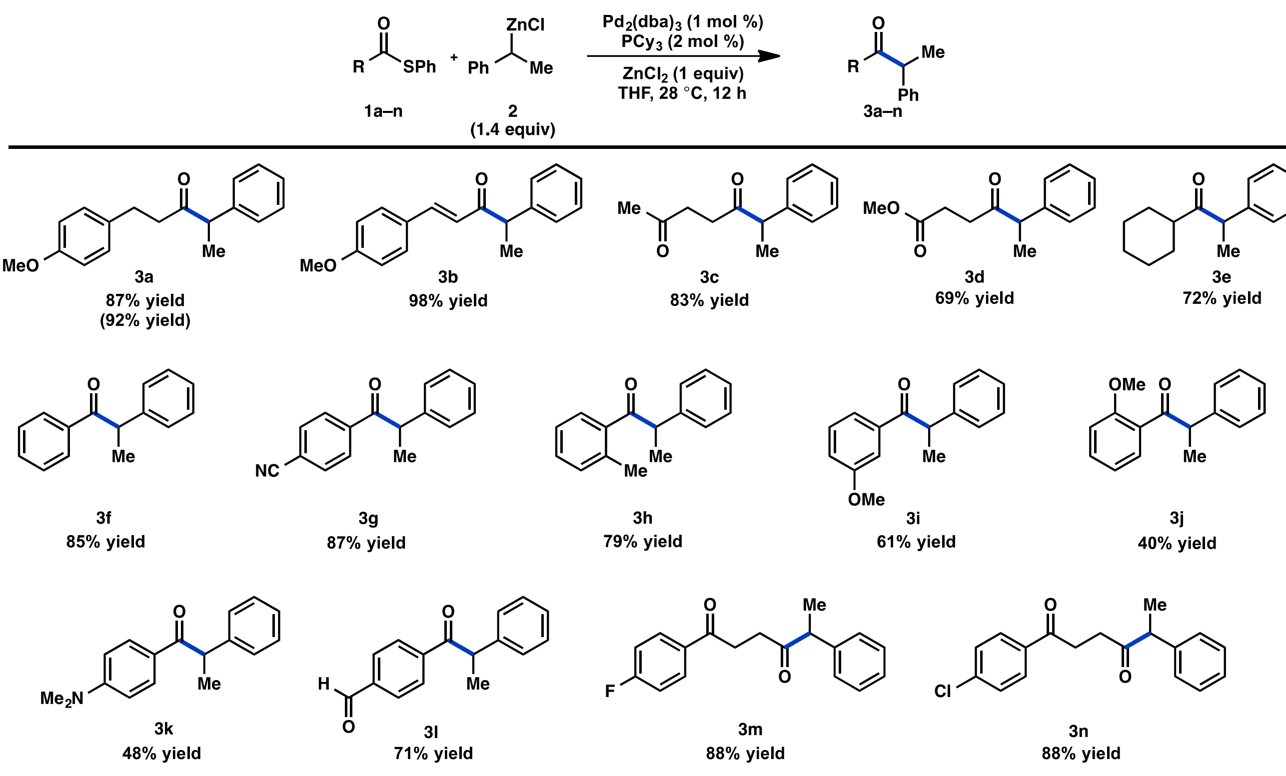
Isolated yields; reactions conducted on a 0.2 mmol scale under an N_2 atmosphere in a glovebox.

Scheme 2. Organozinc substrate scope.

Having developed optimized coupling conditions for secondary organozinc halides, we applied our $Pd_2(dba)_3$/PCy_3/$ZnCl_2$ system to a series of thioesters to determine the scope of the reaction (Scheme 1). The cross-coupling reaction proceeded in excellent yields for alkyl (**1a**), aryl (**1f–l**), and α,β-unsaturated thioesters (**1b**). A variety of thioesters with electrophilic functional groups, such as a ketone (**1c**, **1m**, **1n**), aldehyde (**1l**), ester (**1d**), or nitrile (**1g**), all of which would be incompatible with standard Weinreb ketone synthesis conditions, generate the desired coupling products in high yields. Aryl fluorides (**1m**) and chlorides (**1n**) both react with high chemoselectivity in favor of the thioester. The reaction can also tolerate a more sterically encumbered thioester to form **3h** in 79% yield. In contrast, a decreased yield was observed when a thioester containing an *o*-methoxy group was employed (**1j**). α,α-Disubstituted thioester **1e** furnishes the desired product in 72% yield despite the increased steric hindrance. To illustrate the utility of the methodology, the cross-coupling reaction was performed on a 1.0 mmol scale on the benchtop using standard Schlenk techniques, under which conditions **3a** was obtained in 92% yield.

Given that organozinc **2** was prepared via direct insertion of Zn^0 into (1-chloroethyl)benzene (see Experimental section), we sought to test whether commercial Rieke organozinc reagents,[20] possessing a different cocktail of solubilized salts, would be equally active under our cross-coupling conditions. We were pleased to observe that ketone **3a** could be prepared in 93% yield when using Rieke organozinc reagent **4a** (Scheme 2). Non-activated cyclic and acyclic Rieke organozinc bromides furnished the desired products in excellent yields. In the case of **3o** and **3p**, no isomerized linear product was observed.

While thioesters are air- and moisture-stable building blocks, acid chlorides represent one of the most commonly used types of acyl electrophile. As such, our optimized conditions were tested against several acid chloride coupling partners. Alkyl acid chlorides reacted slowly under the optimized conditions and underwent a competitive Lewis acid-mediated reaction with THF.[21] Aryl acid chlorides proved to be more reactive: in the absence of a $ZnCl_2$

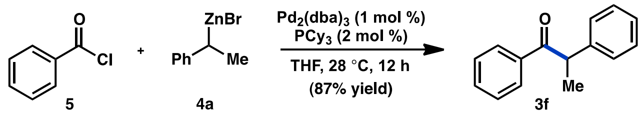


Isolated yields; reactions conducted on a 0.2 mmol scale under an N_2 atmosphere in a glovebox. Number in parentheses indicates reaction conducted on a 1.0 mmol scale under an N_2 atmosphere on the benchtop using standard Schlenk techniques.

Scheme 1. Thioester substrate scope.

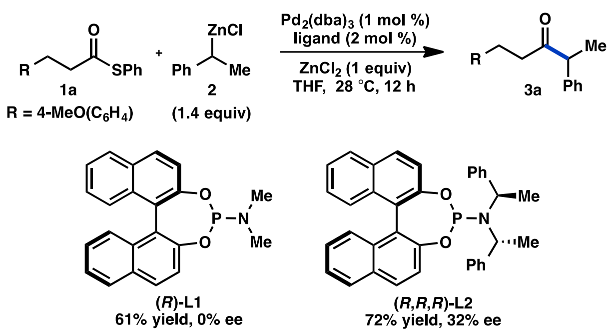
additive, benzoyl chloride (**5**) could be coupled to generate **3f** in 87% yield (Scheme 3).

Having identified conditions for the coupling between thio-



Scheme 3. Extension to benzoyl chloride.

esters and secondary organozinc reagents, we performed preliminary studies to investigate whether the coupling between **2** and thioester **1a** would be amenable to asymmetric catalysis. Kumada and Hayashi exploited the configurational lability of the C–Mg and C–Zn bonds in the Pd-catalyzed enantioselective cross-coupling reaction between (1-phenylethyl) organometallic reagents and vinyl bromides.[22] Knochel has since extended the dynamic kinetic resolution (DKR) strategy to several diastereoselective Negishi cross-couplings of cycloalkylzinc reagents.[23] In agreement with our earlier findings, an assay of chiral phosphine ligands revealed that bidentate phosphines work poorly in the Fukuyama coupling. In contrast, the chiral, monodentate phosphoramidite Monophos (**L1**) provided a moderate yield of **3a**, albeit with no enantioinduction (Scheme 4). A broader analysis of monodentate phosphoramidites showed that **L2** provides the desired ketone with promising levels of enantioselectivity.[24]



Yield of **3a** was calculated by ^1H NMR analysis with internal standard. % ee determined by SFC using a chiral stationary phase.

Scheme 4. Investigation of an asymmetric Fukuyama coupling.

3. Conclusion

In conclusion, a practical method for the Pd-catalyzed Fukuyama cross-coupling of thioesters and secondary organozinc reagents has been developed. The $Pd_2(dba)_3/PCy_3/ZnCl_2$ system furnishes high yields of complex ketone products in a convergent fashion. These conditions also allow for the Negishi coupling of aryl acid chlorides to proceed in high yield. An analysis of chiral ligands revealed that a moderate dynamic kinetic resolution of the organozinc reagent could be achieved, leading to modestly enantioenriched ketone products. Efforts to increase the synthetic utility of the Fukuyama coupling and further extension to asymmetric catalysis are ongoing in our laboratory.

4. Experimental section

4.1. General

Unless otherwise stated, reactions were performed under a nitrogen atmosphere using freshly dried solvents. Tetrahydrofuran (THF) and methylene chloride (CH_2Cl_2) were dried by passing through activated alumina columns. Triethylamine (Et_3N) was distilled over calcium hydride prior to use. Unless otherwise stated, chemicals and reagents were used as received. $Pd_2(dba)_3$ and PCy_3 were purchased from Aldrich and used as received. Phosphoramidite ligands were either purchased from Aldrich or synthesized according to literature procedures. All reactions were monitored by thin-layer chromatography using EMD/Merck silica gel 60 F_{254} pre-coated plates (0.25 mm) and were visualized by UV, *p*-anisaldehyde, or $KMnO_4$ staining. Flash column chromatography was performed as described by Still and co-workers[25] using silica gel (particle size 0.032–0.063) purchased from Silicycle. ^1H and ^{13}C NMR spectra were recorded on a Varian 400 MR (at 400 MHz and 101 MHz, respectively) or a Varian Inova 500 (at 500 MHz and 126 MHz, respectively), and are reported relative to internal chloroform (^1H, δ=7.26, ^{13}C, δ=77.0). Data for ^1H NMR spectra are reported as follows: chemical shift (δ parts per million) (multiplicity, coupling constant (Hertz), integration). Multiplicity and qualifier abbreviations are as follows: s=singlet, d=doublet, t=triplet, q=quartet, m=multiplet, br=broad. IR spectra were recorded on a Perkin Elmer Paragon 1000 spectrometer and are reported in frequency of absorption (cm^{-1}). Analytical SFC was performed with a Mettler SFC supercritical CO_2 analytical chromatography system with Chiralcel AD-H, OD-H, AS-H, OB-H, and OJ-H columns (4.6 mm×25 cm). HRMS were acquired using an Agilent 6200 Series TOF with an Agilent G1978A Multimode source in electrospray ionization (ESI), atmospheric pressure chemical ionization (APCI), or mixed (MM) ionization mode.

4.2. General procedure 1 for synthesis of thioesters

According to the protocol of Tanabe and co-workers,[26] a flask was charged with 3-(4-methoxyphenyl)propionic acid (58.2 mmol, 1 equiv) followed by MeCN (60 mL) and *N*-methyl imidazole (174.6 mmol, 3 equiv). The solution was cooled to 0 °C and tosyl chloride (1.5 M solution in MeCN, 87.3 mmol, 1.5 equiv) was added. The reaction was stirred at 0 °C under N_2 for 0.5 h, after which time thiophenol (1.0 M solution in MeCN, 58.2 mmol, 1 equiv) was added. The reaction was stirred at 0 °C under N_2 for 2 h and then allowed to warm to room temperature. The resulting solution was diluted with ether and water. The aqueous layer was extracted with ether and the combined organic layers were washed with water, dried ($MgSO_4$), filtered, and concentrated. The crude residue was purified by silica gel chromatography.

*4.2.1. S-Phenyl 3-(4-methoxyphenyl)propanethioate (**1a**).* Prepared from 3-(4-methoxyphenyl)propionic acid (10.5 g, 58.2 mmol) following general procedure 1. The crude residue was purified by silica gel chromatography (2:98 to 10:90 EtOAc/hexanes) to yield 13.19 g (83% yield) of **1a** as a white solid, mp 31–32 °C. ^1H NMR (500 MHz, $CDCl_3$) δ 7.47–7.36 (m, 5H), 7.18–7.09 (m, 2H), 6.89–6.81 (m, 2H), 3.80 (s, 3H), 3.03–2.89 (m, 4H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 196.7, 158.1, 134.5, 131.9, 129.3, 129.2, 129.1, 127.6, 113.9, 55.2, 45.4, 30.5; IR (NaCl/thin film): 2932, 2834, 1706, 1611, 1512, 1477, 1440, 1247 cm^{-1}; HRMS (MM) calcd for $[M-H]^-$ 271.0798, found 271.0799.

*4.2.2. Methyl 4-oxo-4-(phenylthio)butanoate (**1d**).* Prepared from monomethyl succinate (2.56 g, 19.4 mmol) following general procedure 1. The crude residue was purified by silica gel chromatography (2:98 to 20:80 EtOAc/hexanes) to yield 2.21 g (51% yield) of **1d** as a slightly yellow oil. ^1H NMR (500 MHz, $CDCl_3$) δ 7.46–7.36 (m, 5H), 3.69 (s, 3H), 3.00 (t, J=6.9 Hz, 2H), 2.69 (t, J=6.9 Hz, 2H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 196.14, 172.2, 134.5, 129.5, 129.2, 127.2, 51.9, 38.0, 28.9; IR (NaCl/thin film): 3061, 2997, 2952, 1739, 1707, 1478, 1440 cm^{-1}; HRMS (MM) calcd for $[M-H]^-$ 223.0434, found 223.0285.

*4.2.3. S-Phenyl 3-methoxybenzothioate (**1i**).* Prepared from 3-methoxybenzoic acid (2.95 g, 19.4 mmol) following general procedure 1. The crude residue was purified by silica gel chromatography (2:98 to 20:80 EtOAc/hexanes) to yield 2.41 g (51% yield) of **1i** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.70–7.63 (m, 1H), 7.57–7.35 (m, 7H), 7.16 (ddd, J=8.2, 2.6, 0.9 Hz, 1H), 3.87 (s, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 190.1, 159.8, 137.9, 135.0, 129.7, 129.5, 129.3, 129.2, 127.3, 120.0, 111.7, 55.5; IR (NaCl/thin film): 3060, 3004, 2938, 2835, 1679, 1597, 1583, 1484, 1440, 1288, 1260 cm^{-1}; HRMS (MM) calcd for $[M-H]^-$ 242.0485, found 242.0462.

*4.2.4. S-Phenyl 4-formylbenzothioate (**1l**).* Prepared from 4-formylbenzoic acid (870 mg, 5.8 mmol) following general procedure 1. The crude residue was purified by silica gel chromatography (2:98 EtOAc/hexanes) to yield 88 mg (6% yield) of **1l** as a white solid. ^{1}H NMR (500 MHz, $CDCl_3$) δ 10.12 (s, 1H), 8.22–8.12 (m, 2H), 8.06–7.95 (m, 2H), 7.58–7.43 (m, 5H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 191.4, 189.6, 141.0, 139.4, 134.9, 129.9 (2C), 129.4, 128.0, 126.6; IR (NaCl/thin film): 3064, 2844, 2744, 1700, 1676, 1575, 1438, 1383, 1198 cm^{-1}; HRMS (MM) calcd for $[M-CHO]^-$ 213.0374, found 213.0373.

4.3. General procedure 2 for synthesis of thioesters

A flame-dried flask was charged with 4-cyanobenzoic acid (5.8 mmol, 1 equiv) and CH_2Cl_2 (12 mL). The reaction was cooled to 0 °C and isobutyl chloroformate (6.38 mmol, 1.1 equiv) and Et_3N (5.8 mmol, 1 equiv) were added dropwise. The resulting mixture was stirred vigorously for 10 min under N_2, after which time Et_3N (5.8 mmol, 1 equiv) and thiophenol (12.76 mmol, 2.2 equiv) were added dropwise. The reaction was stirred at 0 °C under N_2 for 1 h. The reaction was warmed to room temperature and washed with 1 M HCl, water, 1 M NaOH, water, and brine. The combined aqueous layers were extracted with CH_2Cl_2. The combined organic layers were dried ($MgSO_4$), filtered, and concentrated. The crude residue was purified by column chromatography.

*4.3.1. S-Phenyl 4-cyanobenzothioate (**1g**).* Prepared from 4-cyanobenzoic acid (859 mg, 5.8 mmol) following general procedure 2. The crude residue was recrystallized from ether to yield 991 mg (71% yield) of **1g** as a slightly yellow solid, mp 127–129 °C. ^{1}H NMR (500 MHz, $CDCl_3$) 8.17–8.06 (m, 2H), 7.84–7.76 (m, 2H), 7.57–7.44 (m, 5H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 189.1, 139.8, 134.9, 132.6, 130.0, 129.5, 127.9, 126.2, 117.8, 116.9; IR (NaCl/thin film): 3089, 2231, 1683, 1668, 1479, 1404, 1207, 1172 cm^{-1}; HRMS (MM) calcd for $[M-H]^-$ 238.0332, found 238.0327.

*4.3.2. S-Phenyl 4-(4-fluorophenyl)-4-oxobutanethioate (**1m**).* Prepared from 3-(4-fluorobenzoyl)propionic acid (1.14 g, 5.8 mmol) following general procedure 2. The crude residue was purified by silica gel chromatography (2:98 to 20:80 EtOAc/hexanes) to yield 1.44 g (86% yield) of **1m** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 8.05–7.94 (m, 2H), 7.49–7.35 (m, 5H), 7.19–7.07 (m, 2H), 3.34 (t, J=6.6 Hz, 2H), 3.14 (t, J=6.6 Hz, 2H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 196.8, 195.9, 165.8 (d, J_{C-F}=254.8 Hz), 134.5, 132.8, 130.7 (d, J_{C-F}=9.0 Hz), 129.4, 129.2, 127.5, 115.7 (d, J_{C-F}=21.8 Hz), 37.3, 33.3; IR (NaCl/thin film): 3075, 2915, 1705, 1687, 1597, 1506, 1478, 1441, 1409, 1234 cm^{-1}; HRMS (MM) calcd for $[M-H]^-$ 287.0548, found 287.0542.

*4.3.3. S-Phenyl 4-(4-chlorophenyl)-4-oxobutanethioate (**1n**).* Prepared from 3-(4-chlorobenzoyl)propionic acid (1.23 g, 5.8 mmol) following general procedure 2. The crude residue was purified by silica gel chromatography (2:98 to 20:80 EtOAc/hexanes) to yield 1.08 g (61% yield) of **1n** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.97–7.85 (m, 2H), 7.48–7.35 (m, 7H), 3.33 (t, J=6.6 Hz, 2H), 3.14 (t, J=6.5 Hz, 2H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 196.7, 196.3, 139.7, 134.6, 134.5, 129.4, 129.3, 129.2, 128.9, 127.4, 37.2, 33.4; IR (NaCl/thin film): 3060, 2914, 1706, 1685, 1590, 1478, 1441, 1400 cm^{-1}; HRMS (MM) calcd for $[M+H]^+$ 305.0398, found 305.0406.

4.4. Preparation of organozinc 2

According to the protocol of Orito and co-workers,[27] a flame-dried Schlenk flask was charged with zinc dust (2.80 g, 42.6 mmol, 1.5 equiv), TMSCl (0.14 mL, 1.12 mmol, 0.04 equiv), and THF (14 mL). The reaction was stirred at room temperature under N_2 for 15 min. To the reaction was added (1-chloroethyl)benzene (4.00 g, 28.4 mmol, 1 equiv) dropwise. The reaction was stirred at 40 °C under N_2 for 4 h. The mixture was cooled to room temperature and the excess zinc dust was allowed to settle. The solution was titrated with I_2 according to Knochel and co-workers.[28] The yellow organozinc solution was transferred to a flame-dried flask and stored under N_2 at −20 °C. Concentrations of 1.0 M were obtained.

4.5. General procedure for Fukuyama cross-coupling reaction

In a glovebox, a vial was charged with $Pd_2(dba)_3$ (0.002 mmol, 0.01 equiv), PCy_3 (0.004 mmol, 0.02 equiv), and THF (1 mL). The deep purple solution was stirred at 28 °C for 10 min. The thioester substrate (0.2 mmol, 1 equiv), the (1-phenylethyl)zinc(II) chloride (**2**, 1.0 M THF, 0.28 mmol, 1.4 equiv), and $ZnCl_2$ (1.0 M THF, 0.2 mmol, 1 equiv) were added to the reaction. The reaction was stirred at 28 °C for 12 h, after which time it was removed from the glovebox, quenched with 0.1 mL satd NH_4Cl, dried ($MgSO_4$), filtered, and concentrated. The crude residue was purified by silica gel chromatography.

*4.5.1. 1-(4-Methoxyphenyl)-4-phenylpentan-3-one (**3a**).* Prepared from *S*-phenyl 3-(4-methoxyphenyl)propanethioate (**1a**, 54.1 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 to 20:80 EtOAc/hexanes) to yield 46.8 mg (87% yield) of **3a** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.37–7.21 (m, 3H), 7.22–7.14 (m, 2H), 7.05–6.96 (m, 2H), 6.84–6.75 (m, 2H), 3.79 (s, 3H), 3.72 (q, J=7.0 Hz, 1H), 2.88–2.57 (m, 4H), 1.39 (d, J=7.0 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 210.0, 157.8, 140.4, 133.0, 129.2, 128.9, 127.8, 127.1, 113.7, 55.2, 53.2, 42.8, 29.1, 17.3; IR (NaCl/thin film): 3060, 3027, 2973, 2931, 2834, 1713, 1611, 1513, 1493, 1452, 1300, 1247 cm^{-1}; HRMS (MM) calcd for $[M-H]^-$ 267.1391, found 267.1391.

*4.5.2. (E)-1-(4-Methoxyphenyl)-4-Phenylpent-1-en-3-one (**3b**).* Prepared from (*E*)-*S*-phenyl 3-(4-methoxyphenyl)prop-2-enethioate (**1b**, 54.1 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 to 20:80 EtOAc/hexanes) to yield 52.0 mg (98% yield) of **3b** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.59 (d, J=15.9 Hz, 1H), 7.47–7.39 (m, 2H), 7.37–7.19 (m, 5H), 6.91–6.80 (m, 2H), 6.60 (d, J=15.9 Hz, 1H), 4.01 (q, J=6.9 Hz, 1H), 3.81 (s, 3H), 1.49 (d, J=6.9 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 199.4, 161.4, 142.4, 140.9, 130.0, 128.9, 128.0, 127.2, 127.0, 122.4, 114.2, 55.3, 51.7, 17.9; IR (NaCl/thin film): 2971, 2930, 2837, 1684, 1654, 1598, 1572, 1511, 1254 cm^{-1}; HRMS (MM) calcd for $[M-H]-$ 265.1234, found 265.1239.

*4.5.3. Methyl 4-oxo-5-phenylhexanoate (**3d**).* Prepared from methyl 4-oxo-4-(phenylthio)butanoate (**1d**, 44.9 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 to 20:80 EtOAc/hexanes) to yield 30.3 mg (69% yield) of **3d** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.39–7.16 (m, 5H), 3.79 (q,

J=7.0 Hz, 1H), 3.63 (s, 3H), 2.77–2.38 (m, 4H), 1.41 (d, J=7.0 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 209.0, 173.2, 140.4, 128.9, 127.9, 127.2, 52.9, 51.7, 35.6, 27.9, 17.3; IR (NaCl/thin film): 3062, 3027, 2977, 2952, 2932, 1739, 1716, 1494, 1453, 1437, 1372, 1356, 1211 cm^{-1}; HRMS (MM) calcd for [M+H]$^+$ 221.1172, found 221.1173.

*4.5.4. 4-(2-Phenylpropanoyl)benzonitrile (**3g**).* Prepared from *S*-phenyl 4-cyanobenzothioate (**1g**, 47.9 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 to 20:80 EtOAc/hexanes) to yield 40.8 mg (87% yield) of **3g** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 8.04–7.94 (m, 2H), 7.71–7.61 (m, 2H), 7.39–7.17 (m, 5H), 4.61 (q, J=6.8 Hz, 1H), 1.54 (d, J=6.8 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 198.8, 140.4, 139.5, 132.3, 129.3, 129.1, 127.6, 127.3, 117.9, 115.9, 48.6, 19.3; IR (NaCl/thin film): 3061, 2976, 2930, 2230, 1685, 1600, 1452, 1404, 1218, 1199 cm^{-1}; HRMS (MM) calcd for [M−H]$^-$ 234.0924, found 234.0932.

*4.5.5. 2-Phenyl-1-(o-tolyl)propan-1-one (**3h**).* Prepared from *S*-phenyl 2-methylbenzothioate (**1h**, 45.7 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 EtOAc/hexanes) to yield 35.6 mg (79% yield) of **3h** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.53 (d, J=7.7 Hz, 1H), 7.36–7.06 (m, 8H), 4.53 (q, J=6.9 Hz, 1H), 2.34 (s, 3H), 1.55 (d, J=6.9 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) 204.6, 140.4, 138.5, 137.8, 131.5, 130.7, 128.8, 127.9, 127.8, 126.9, 125.3, 50.7, 20.8, 18.5; IR (NaCl/thin film): 3061, 3026, 2971, 2929, 2870, 1685, 1600, 1492, 1452, 1220 cm^{-1}; HRMS (MM) calcd for [M−H]$^-$ 238.1128, found 238.1128.

*4.5.6. 1-(3-Methoxyphenyl)-2-phenylpropan-1-one (**3i**).* Prepared from *S*-phenyl 3-methoxybenzothioate (**1i**, 48.9 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 EtOAc/hexanes) to yield 29.2 mg (61% yield) of **3i** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.56–7.44 (m, 2H), 7.34–7.25 (m, 5H), 7.24–7.15 (m, 1H), 7.02 (ddd, J=8.2, 2.7, 0.9 Hz, 1H), 4.67 (q, J=6.9 Hz, 1H), 3.80 (s, 3H), 1.53 (d, J=6.8 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 200.1, 159.7, 141.5, 137.8, 129.4, 129.0, 127.7, 126.9, 121.4, 119.3, 113.1, 55.3, 48.0, 19.5; IR (NaCl/thin film): 3062, 3025, 2973, 2930, 2835, 1681, 1596, 1581, 1488, 1451, 1426, 1263 cm^{-1}; HRMS (MM) calcd for [M−H]$^-$ 239.1078, found 239.1088.

*4.5.7. 1-(2-Methoxyphenyl)-2-phenylpropan-1-one (**3j**).* Prepared from *S*-phenyl 2-methoxybenzothioate (**1j**, 48.9 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 EtOAc/hexanes) to yield 19.2 mg (40% yield) of **3j** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.48 (dd, J=7.6, 1.8 Hz, 1H), 7.39 (ddd, J=8.3, 7.3, 1.8 Hz, 1H), 7.34–7.13 (m, 5H), 6.95–6.89 (m, 2H), 4.75 (q, J=7.0 Hz, 1H), 3.88 (s, 3H), 1.54 (d, J=6.9 Hz, 5H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 204.1, 157.5, 141.3, 135.1, 132.7, 130.4, 128.4, 128.1, 126.6, 120.5, 111.3, 55.3, 51.8, 18.7; IR (NaCl/thin film): 2929, 1736, 1666, 1597, 1485, 1466, 1284, 1245, 1198 cm^{-1}; HRMS (MM) calcd for [M+H]$^+$ 241.1223, found 241.1223.

*4.5.8. 1-(4-(Dimethylamino)phenyl)-2-phenylpropan-1-one (**3k**).* Prepared from *S*-phenyl 4-(dimethylamino)benzothioate (**1k**, 51.5 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 to 20:80 EtOAc/hexanes) to yield 24.4 mg (48% yield) of **3k** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.95–7.85 (m, 2H), 7.36–7.23 (m, 4H), 7.23–7.13 (m, 1H), 6.62–6.55 (m, 2H), 4.64 (q, J=6.9 Hz, 1H), 3.00 (s, 6H), 1.51 (d, J=6.9 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 198.3, 153.1, 142.6, 131.0, 130.9, 128.7, 126.5, 124.3, 110.6, 46.9, 39.9, 19.6; IR (NaCl/thin film): 3027, 2974, 2928, 2820, 1651, 1615, 1492, 1448, 1234, 1198 cm^{-1}; HRMS (MM) calcd for [M+H]$^+$ 254.1539, found 254.1543.

*4.5.9. 4-(2-Phenylpropanoyl)benzaldehyde (**3l**).* Prepared from *S*-phenyl 4-formylbenzothioate (**1l**, 48.5 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 to 20:80 EtOAc/hexanes) to yield 33.7 mg (71% yield) of **3l** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 10.03 (s, 1H), 8.11–7.98 (m, 2H), 7.95–7.75 (m, 2H), 7.45–7.12 (m, 5H), 4.67 (q, J=6.8 Hz, 1H), 1.55 (d, J=6.9 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 199.7, 191.5, 140.7, 138.6, 129.7, 129.2, 128.9, 127.7, 127.2, 126.8, 48.6, 19.3; IR (NaCl/thin film): 3060, 3026, 2975, 2929, 1700, 1684, 1606, 1452, 1219 cm^{-1}; HRMS (MM) calcd for [M−H]$^-$ 237.0921, found 237.0909.

*4.5.10. 1-(4-Fluorophenyl)-5-phenylhexane-1,4-dione (**3m**).* Prepared from *S*-phenyl 4-(4-fluorophenyl)-4-oxobutanethioate (**1m**, 57.7 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 to 10:90 EtOAc/hexanes) to yield 50.3 mg (88% yield) of **3m** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 8.03–7.92 (m, 2H), 7.41–7.33 (m, 2H), 7.30–7.21 (m, 3H), 7.16–7.05 (m, 2H), 3.90 (q, J=7.0 Hz, 1H), 3.28 (ddd, J=18.1, 7.6, 5.8 Hz, 1H), 3.06 (dt, J=18.1, 6.0 Hz, 1H), 2.88 (ddd, J=18.2, 7.6, 5.7 Hz, 1H), 2.72 (dt, J=18.2, 6.0 Hz, 1H), 1.44 (d, J=7.0 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 209.5, 197.0, 165.7 (d, J_{C-F}=254.4 Hz), 140.6, 133.1, 130.7 (d, J_{C-F}=9.1 Hz), 128.9, 127.9, 127.2, 115.6 (d, J_{C-F}=21.8 Hz), 53.1, 34.8, 32.5, 17.4; IR (NaCl/thin film): 2975, 2913, 1714, 1685, 1597, 1506, 1410, 1231 cm^{-1}; HRMS (MM) calcd for [M−H]$^-$ 283.1140, found 283.1141.

*4.5.11. 1-(4-Chlorophenyl)-5-phenylhexane-1,4-dione (**3n**).* Prepared from *S*-phenyl 4-(4-chlorophenyl)-4-oxobutanethioate (**1n**, 61.0 mg, 0.2 mmol). The crude residue was purified by silica gel chromatography (2:98 to 10:90 EtOAc/hexanes) to yield 52.8 mg (88% yield) of **3n** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.91–7.83 (m, 2H), 7.48–7.22 (m, 7H), 3.89 (q, J=7.0 Hz, 1H), 3.26 (ddd, J=18.1, 7.5, 5.8 Hz, 1H), 3.04 (dt, J=18.2, 5.9 Hz, 1H), 2.87 (ddd, J=18.2, 7.6, 5.6 Hz, 1H), 2.72 (dt, J=18.2, 6.0 Hz, 1H), 1.44 (d, J=7.0 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 209.5, 197.4, 140.6, 139.5, 134.9, 129.4, 128.9, 128.8, 127.9, 127.2, 53.1, 34.8, 32.5, 17.4; IR (NaCl/thin film): 3062, 3026, 2974, 2913, 1713, 1685, 1590, 1493, 1452, 1400, 1199 cm^{-1}; HRMS (MM) calcd for [M−H]$^-$ 299.0844, found 299.0870.

*4.5.12. 1-(4-Methoxyphenyl)-4-methylhexan-3-one (**3o**).* Prepared from *S*-phenyl 3-(4-methoxyphenyl)propanethioate (**1a**, 54.5 mg, 0.2 mmol) and Rieke organozinc **4b** (0.5 M THF, 0.56 mL, 0.28 mmol). The crude residue was purified by silica gel chromatography (5:95 EtOAc/hexanes) to yield 41.4 mg (94% yield) of **3o** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.15–7.05 (m, 2H), 6.87–6.76 (m, 2H), 3.78 (s, 3H), 2.83 (t, J=7.2 Hz, 2H), 2.81–2.66 (m, 2H), 2.41 (h, J=6.9 Hz, 1H), 1.71–1.58 (m, 1H), 1.44–1.29 (m, 1H), 1.03 (d, J=6.9 Hz, 3H), 0.83 (t, J=7.4 Hz, 3H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 213.9, 157.3, 133.4, 129.2, 113.8, 55.2, 48.0, 43.0, 28.8, 25.82, 15.7, 11.6; IR (NaCl/thin film): 2964, 2933, 2875, 2835, 1710, 1612, 1513, 1462, 1300, 1247 cm^{-1}; HRMS (MM) calcd for [M]$^+$ 220.1458, found 220.1414.

*4.5.13. 1-(4-Methoxyphenyl)-4-methylpentan-3-one (**3p**).* Prepared from *S*-phenyl 3-(4-methoxyphenyl)propanethioate (**1a**, 54.5 mg, 0.2 mmol) and Rieke organozinc **4c** (0.5 M THF, 0.56 mL, 0.28 mmol). The crude residue was purified by silica gel chromatography (5:95 EtOAc/hexanes) to yield 38.2 mg (92% yield) of **3p** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.13–7.06 (m, 2H), 6.87–6.78 (m, 2H), 3.78 (s, 3H), 2.87–2.78 (m, 2H), 2.78–2.68 (m, 2H), 2.56 (hept, J=7.0 Hz, 1H), 1.06 (d, J=6.9 Hz, 6H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 214.0, 157.8, 133.4, 129.2, 113.8, 55.2, 42.2, 41.0, 28.9, 18.1; IR (NaCl/thin film): 2968, 2933, 2873, 2835, 1710, 1612, 1513, 1465, 1300, 1246 cm^{-1}; HRMS (MM) calcd for [M−H]$^-$ 205.1234, found 205.1575.

*4.5.14. 1-Cyclopentyl-3-(4-methoxyphenyl)propan-1-one (**3q**).* Prepared from *S*-phenyl 3-(4-methoxyphenyl)propanethioate (**1a**, 54.5 mg, 0.2 mmol) and Rieke organozinc **4d** (0.5 M THF, 0.56 mL, 0.28 mmol). The crude residue was purified by silica gel chromatography (5:95 EtOAc/hexanes) to yield 41.0 mg (88% yield) of **3q** as a slightly yellow oil. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.15–7.05 (m, 2H), 6.87–6.78 (m, 2H), 3.78 (s, 3H), 2.88–2.76 (m, 3H), 2.78–2.69 (m, 2H), 1.84–1.49 (m, 8H); ^{13}C NMR (125 MHz, $CDCl_3$) δ 212.4, 157.8, 133.4, 129.2, 113.8, 55.2, 51.5, 43.6, 29.0, 28.7, 25.9; IR (NaCl/thin film): 2953, 2868, 1707, 1612, 1513, 1300, 1247 cm^{-1}; HRMS (MM) calcd for $[M-H]^-$ 233.1536, found 233.1530.

4.6. Procedure for acid chloride cross-coupling reaction

In a glovebox, a vial was charged with $Pd_2(dba)_3$ (0.002 mmol, 0.01 equiv), PCy_3 (0.004 mmol, 0.02 equiv), and THF (1 mL). The deep purple solution was stirred at 28 °C for 10 min. Acid chloride **5** (0.2 mmol, 1 equiv) and Rieke organozinc bromide **4a** (0.5 M THF, 0.56 mmol, 1.4 equiv) were added to the reaction. The reaction was stirred at 28 °C for 12 h, after which time it was removed from the glovebox, quenched with 0.1 mL satd NH_4Cl, dried ($MgSO_4$), filtered, and concentrated. The crude residue was purified by silica gel chromatography.

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